

KANISHK GOEL

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EDUCATION

Bachelor of Technology in Microelectronics and VLSI

2025-2029

IIT Mandi, CGPA: 9.68

COURSES

EE-302 Control systems.

EE-212 Digital system design.

EE-260 Signals and systems.

EE-203 Network Theory

PROJECTS

Memory buffer architecture for AI hardware accelerator chips

- Studied Google TPU V1 architecture for systolic arrays for AI hardware acceleration, and designed a memory buffer architecture with reasoning covering tradeoffs to maximize speed, power efficiency and cost

FOC control algorithm and Gate driver and inverter design

Initially studied the foc control algorithm and the pre-requisite motor control and physics knowledge. Subsequently worked on designing the schematic and pcb for the gate driver and inverter part of the FOC board. Learnt to deal with parasitics, how to minimise them. How to deal with high currents and temperatures, how to decide what via's to use, and where to place components and routes to minimize signal noise. Learnt the basics of analog design and passive component selection process.

Worked on the PS-3 of the electrical team

- Studied different control algorithms, and mathematics of control theory, understood the practical implications of different control algorithms and how to overcome them.